



Part 2 of 2

DESIGNING LOW-POWER PRODUCTS USING MSP430 MICROCONTROLLER

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The first part of this article described the construction of a Flash emulation tool based on the PC's parallel port called 'ParaFET,' for programming the MSP430 devices. MSP430F2013 has been chosen for building this experimental circuit since this chip is available in DIP package. The important features and pin details of MSP430F2013 have already been discussed in the first part.

In continuation, we describe here construction of the target board to test the working of the ParaFET circuit. The internal block diagram of MSP430F2013 is shown in Fig. 9.

Building the applications

The target board combines two applications: a blinking LED and a

precision digital thermometer based on MSP430F2013 chip. The circuit is shown in Fig. 10. (Parts list has already been included in Part 1.)

To make you familiar with MSP430 programming and debugging, let us begin with a simple blinking-LED application. For this purpose, it is recommended to use a 3mm or 5mm LED which consumes less than 2 mA. Resistor R25 and capacitor C3 provide power-on reset to the MSP430. C4 and C5 are filter capacitors between V_{CC} and GND. The connections include the TDO, TDI, TMS, TCK, TEST, RESET, $+V_{CC}$ and GND lines, which are terminated at the 14-pin JTAG connector (or FRC connector).

Software program

The Blinking_LED.c program is written in 'C' language. (It is given at the

end of this article.) The program starts with the msp430x20x3.h header file for MSP430F2013 device. This file is automatically picked up by the compiler from the IAR installation directory during program compilation. At the beginning of the program, the watchdog timer needs to be stopped. Otherwise, it will cause a non-maskable interrupt after about 32 ms.

Now, set up the clock system. By default, MCLK and SMCLK start off with the internal digitally controlled oscillator (DCO) at 1 MHz. ACLK has been programmed to use the internal very low-frequency oscillator (VLO), whose oscillation frequency is approximately 10 kHz. So there is no need to connect any external crystal.

Next, configure the port modules. The unused bits of port 1 are programmed as outputs with a value of

zero. This ensures that stray voltages on this port don't cause inadvertent leakage currents. Port 2.6 is configured as the output to drive the LED through resistor R29.

Programming and debugging

The steps for programming and debugging after installing the IAR Embedded Workbench KickStart for MSP430 in your system follow:

1. Connect the parallel-port FET board (ParaFET) to the PC's parallel port.

2. Connect the

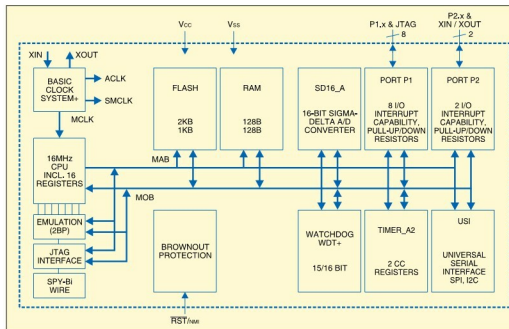


Fig. 9: Internal block diagram of MSP430F2013